

DIANA PRINCE

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Performance-driven Senior Software Engineer with over 5 years of experience architecting high-throughput backend services, intelligent data pipelines, and responsive enterprise interfaces. Proven track record of bridging AI/ML capabilities with rock-solid infrastructure, optimizing low-latency vector databases, and leading agile engineering teams.

TECHNICAL SKILLS

Languages:	Python (Advanced), TypeScript, JavaScript, SQL, C++, Go
Data & AI/ML:	pgvector, Sentence-BERT, Transformers, PyTorch, OCR Engines (Tesseract/PaddleOCR), Semantic Ranking, RAG
Frameworks:	FastAPI, React, PostgreSQL, Django, Node.js, Express, Redux Toolkit, TailwindCSS
Infrastructure:	Scalable Microservices, API Contract Design (OpenAPI), Docker, Kubernetes, AWS (EC2, S3, RDS), CI/CD

PROFESSIONAL EXPERIENCE

Nexus Tech Enterprises <i>Lead Software Engineer — AI Solutions Group</i>	June 2024 – Present New York, NY
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- Architected and scaled an asynchronous document processing and intelligence platform utilizing **Python and FastAPI**, supporting 10M+ daily API requests with 99.9% uptime.
- Designed low-latency vector database infrastructure using **PostgreSQL and pgvector**, optimizing multi-dimensional index scanning to reduce semantic retrieval latency from 240ms to 38ms.
- Deployed and fine-tuned **Sentence-BERT embedding pipelines** for high-accuracy text processing and semantic ranking, achieving a 22% improvement in contextual document matching metrics.
- Led technical design and rigorous code reviews for a team of 6 engineers, establishing robust engineering best practices, strict API contract validations, and automated regression testing suites.
- Spearheaded the integration of open-source **OCR engines** and custom PDF parsing libraries to cleanly ingest and transform highly unstructured, messy data extraction streams.

Quantum Core Systems <i>Senior Full-Stack Engineer</i>	September 2021 – May 2024 Boston, MA
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- Built robust, type-safe developer dashboards and complex administrative web tools using **React and TypeScript**, optimizing rendering performance for heavy real-time data visualization streams.
- Migrated a legacy monolithic system into a scalable microservices architecture, isolating heavy data ingestion workloads and successfully eliminating operational single points of failure.
- Partnered with Product and Data Science teams to transition raw ML model proofs-of-concept into production-ready, horizontally scalable cloud services on AWS infrastructure.
- Mentored 4 junior and mid-level developers, implementing technical pair programming sessions and weekly architecture focus talks that accelerated core team delivery velocity by 15%.

TECHNICAL PROJECTS & CONTRIBUTIONS

DocuMind — Open-Source RAG Framework <i>Core Contributor & Maintainer</i>	January 2025 – Present
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- Engineered an end-to-end open-source framework utilizing **FastAPI** and modern vector stores for rapid Retrieval-Augmented Generation deployments.
- Authored the TypeScript-based **React** component library used by over 1,200 GitHub developers to easily interface with complex vector embeddings.

EDUCATION

Carnegie Mellon University <i>B.S. in Computer Science</i>	Graduation: May 2021 Pittsburgh, PA
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Track Specialties: Distributed Systems, Natural Language Processing, Advanced Database Tuning.